MAR-PE-13 Cisco PE Routing Protocol Flap SOP

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_8316

Description

This Standard Operating Procedure (SOP) outlines the steps to be taken in the event of a major alarm, ALARM_8316, on Cisco PE MAR-PE-13 at the Marathahalli Core site. The failure domain of interest is the routing protocol flap, encompassing BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. The underlying cause of this failure domain can be attributed to the complex interplay of routing protocol timers, convergence mechanisms, and link state management on the PE device. Specifically, the Cisco PE's reliance on BGP's path vector protocol and OSPF's link-state protocol, in conjunction with ISIS's LSP flooding mechanism, creates a fragile equilibrium that can be disrupted by a multitude of factors, including but not limited to, routing table instability, link failures, or misconfigurations. Furthermore, the PE device's role in aggregating and forwarding traffic, while ensuring routing protocol convergence and stability, makes it a critical component in the overall Marathahalli Core network infrastructure.

Pre-requisite Checks

- check: Verify that the Marathahalli Core site is accessible and that all necessary personnel and equipment are on site.

- check: Confirm that the redundant link status is functioning correctly and that there are no signs of impending failure.

- check: Ensure that the necessary permissions and access credentials have been granted to perform the required diagnostic and resolution steps.

Potential Root Causes

Cause	Probability
BGP routing table instability due to misconfiguration or hardware failure.	High
OSPF adjacency loss due to link failure or misconfiguration.	Medium
ISIS LSP flooding due to misconfiguration or hardware failure.	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 10, local router ID is 192.168.1.1, Description: This command displays a summary of the BGP table, including the table version, local router ID, and the number of paths in the table.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays a list of OSPF neighbors, including their ID, priority, state, dead time, address, and interface.
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```
Command: show isis database, Expected Output: IS-IS Level-1 Link-State Database:, Description: This command displays the ISIS database, including the level, link-state IDs, and sequence numbers.
```

Resolution Steps

Step 1: Step: Clear BGP routing table instability, Command: clear ip bgp *, Description: This command clears the BGP routing table, resetting the table to its default state.

Step 2: Step: Establish OSPF adjacency, Command: clear ip ospf process, Description: This command clears the OSPF process, resetting the adjacency state and establishing a new adjacency.

Step 3: Step: Flush ISIS LSP flooding, Command: clear isis database, Description: This command flushes the ISIS LSP flooding, resetting the database to its default state.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 10, local router ID is 192.168.1.1, Description: This command verifies that the BGP table has been cleared and is in a stable state.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command verifies that OSPF adjacency has been established and is stable.

Command: show isis database, Expected Output: IS-IS Level-1 Link-State Database:, Description: This command verifies that the ISIS database has been flushed and is in a stable state.
```

Roll-back Procedures

- Step: Restore BGP routing table, Command: show ip bgp summary, Description: This command restores the BGP routing table to its previous state.

- Step: Re-establish OSPF adjacency, Command: clear ip ospf process, Description: This command re-establishes OSPF adjacency to its previous state.

- Step: Re-flush ISIS LSP flooding, Command: clear isis database, Description: This command re-flushes the ISIS LSP flooding to its previous state.